

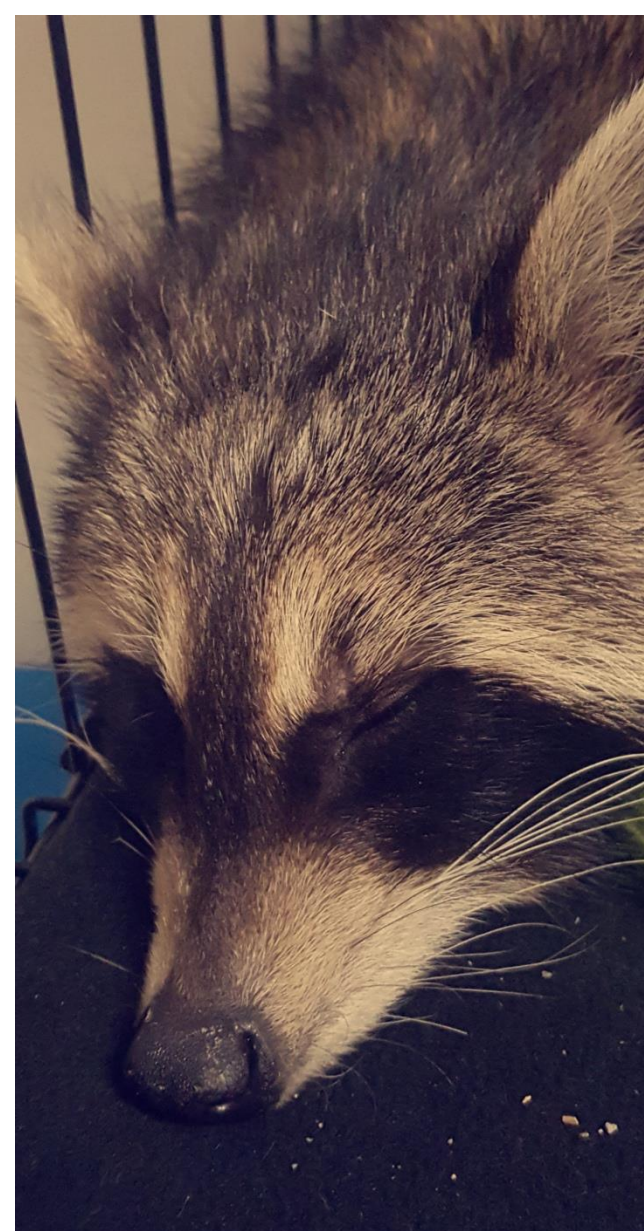


Microscopic Electric Fields of Hot, Dense Plasmas

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ABOUT ME

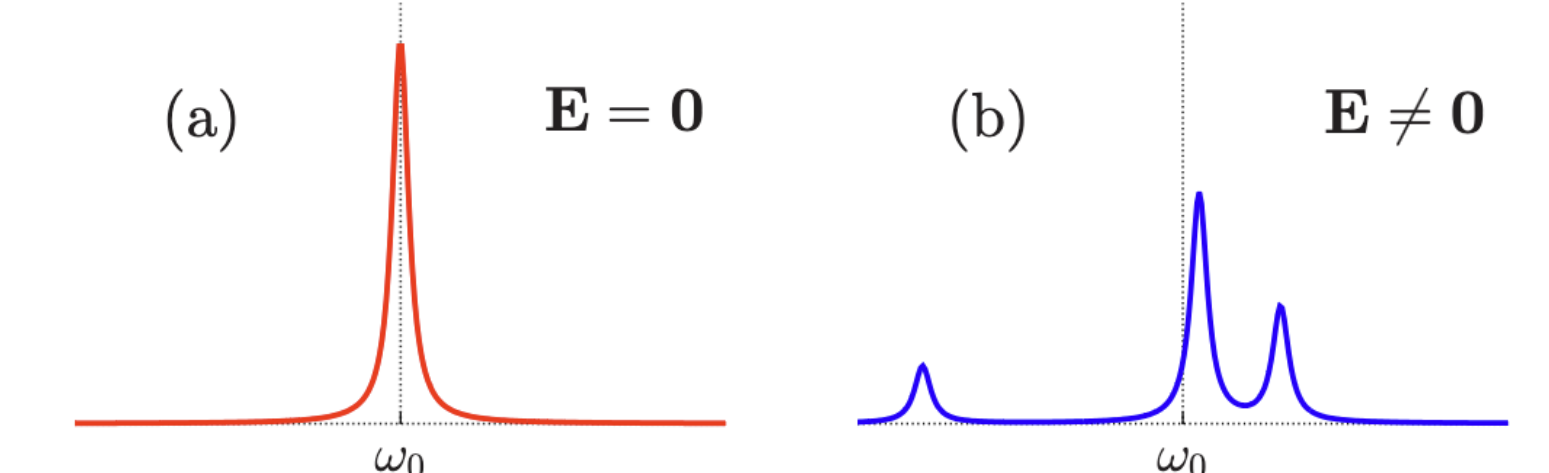
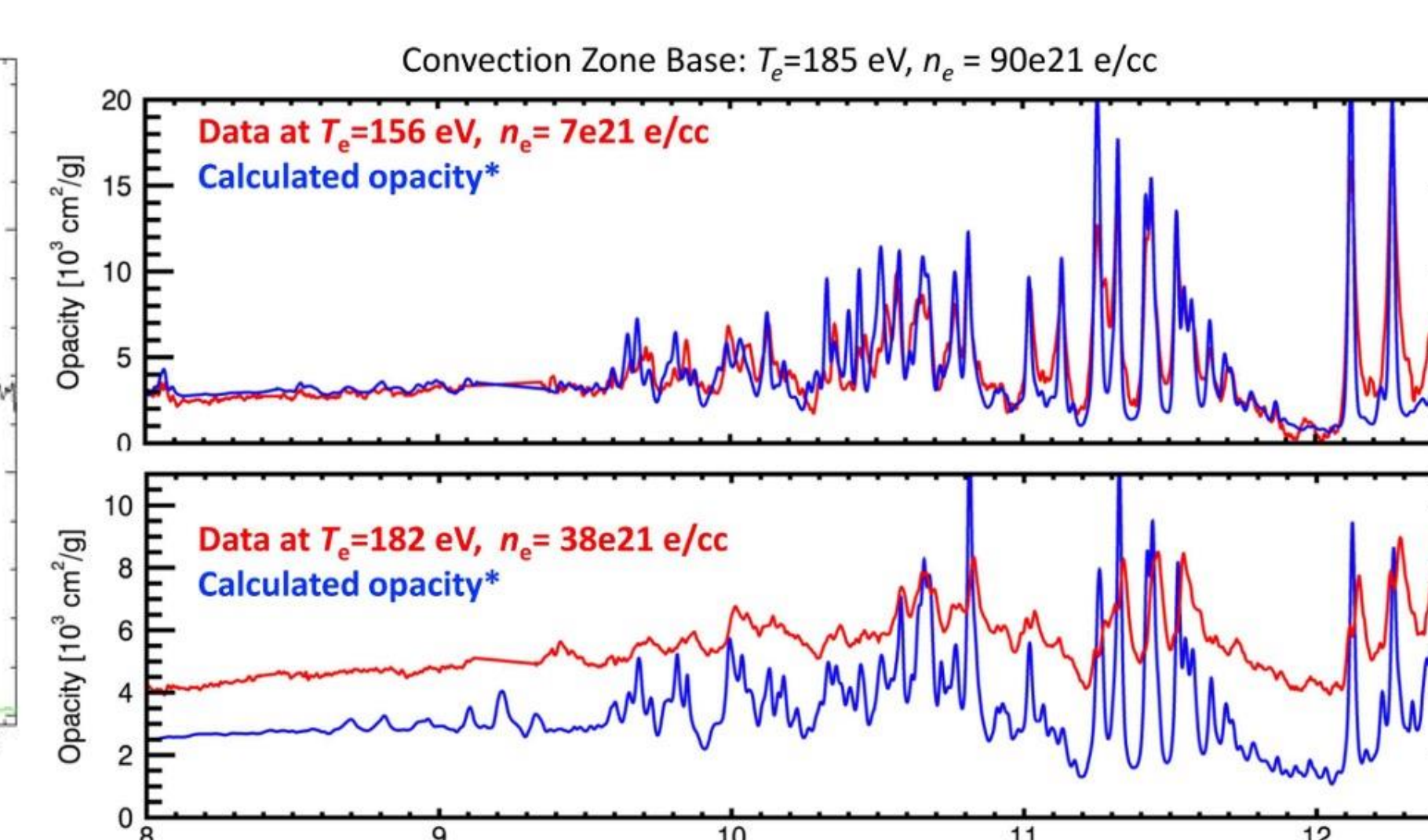
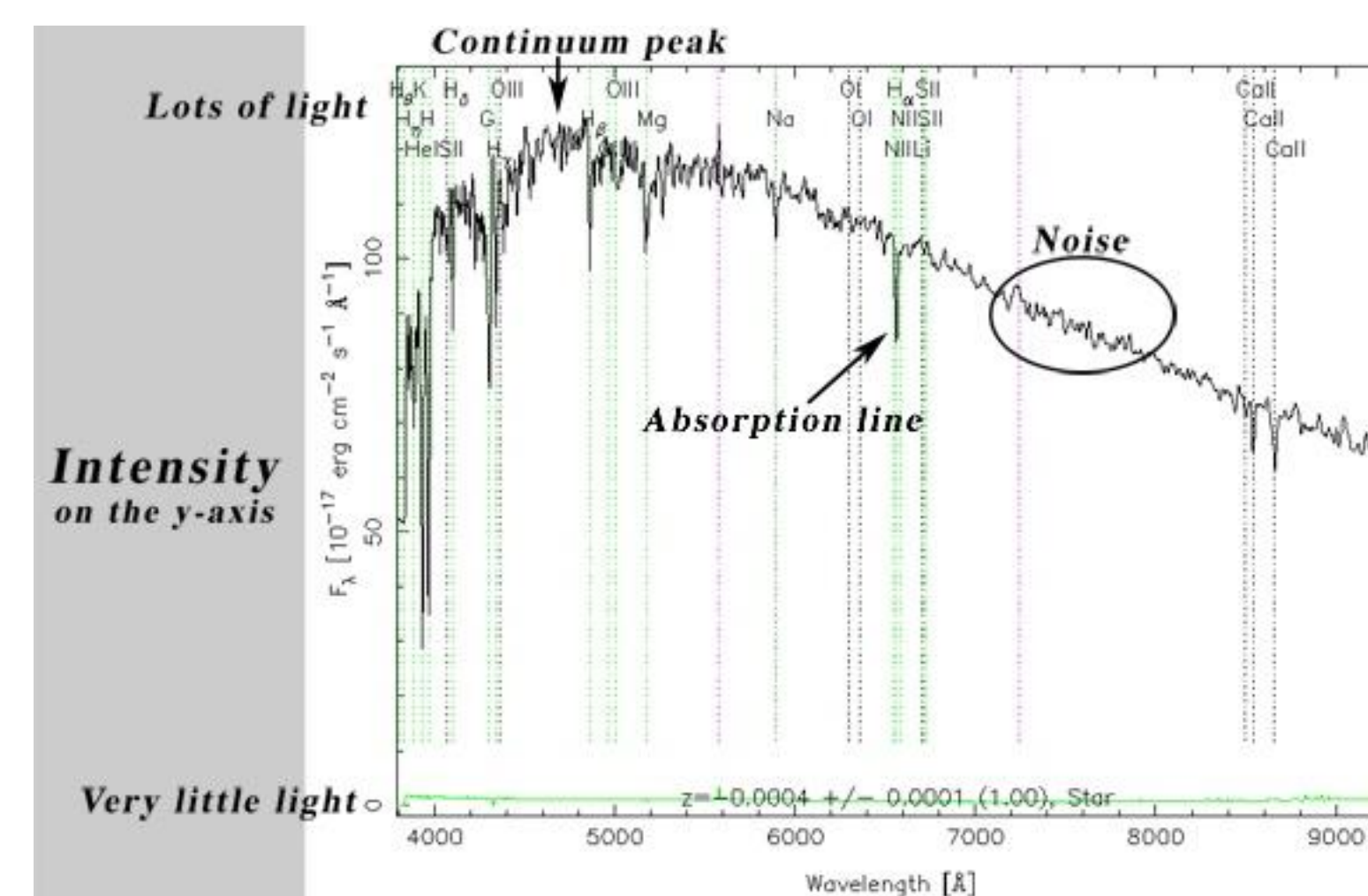
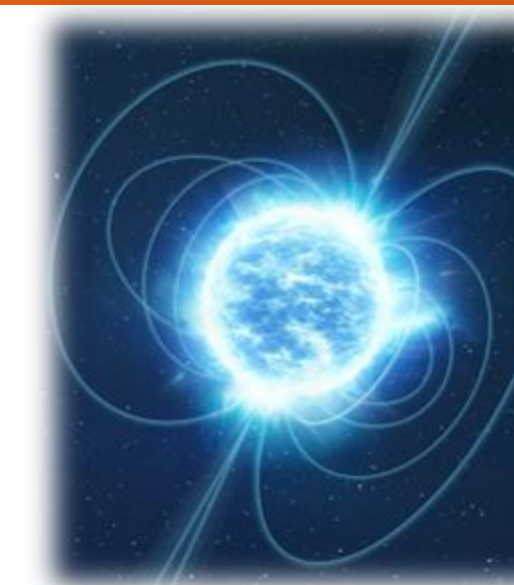
- Astrophysics & Anthropology Bachelor's Recipient; 2nd Year Physics Master's Student
- I aim to work in stellar or compact object modeling as a staff scientist at a national lab.
- I came to LLNL to experience lab work and improve my skills in computational physics. LLF has helped me connect with other students and scientists.
- My interests include but are not limited to kayaking, guitar, and animal rescue.



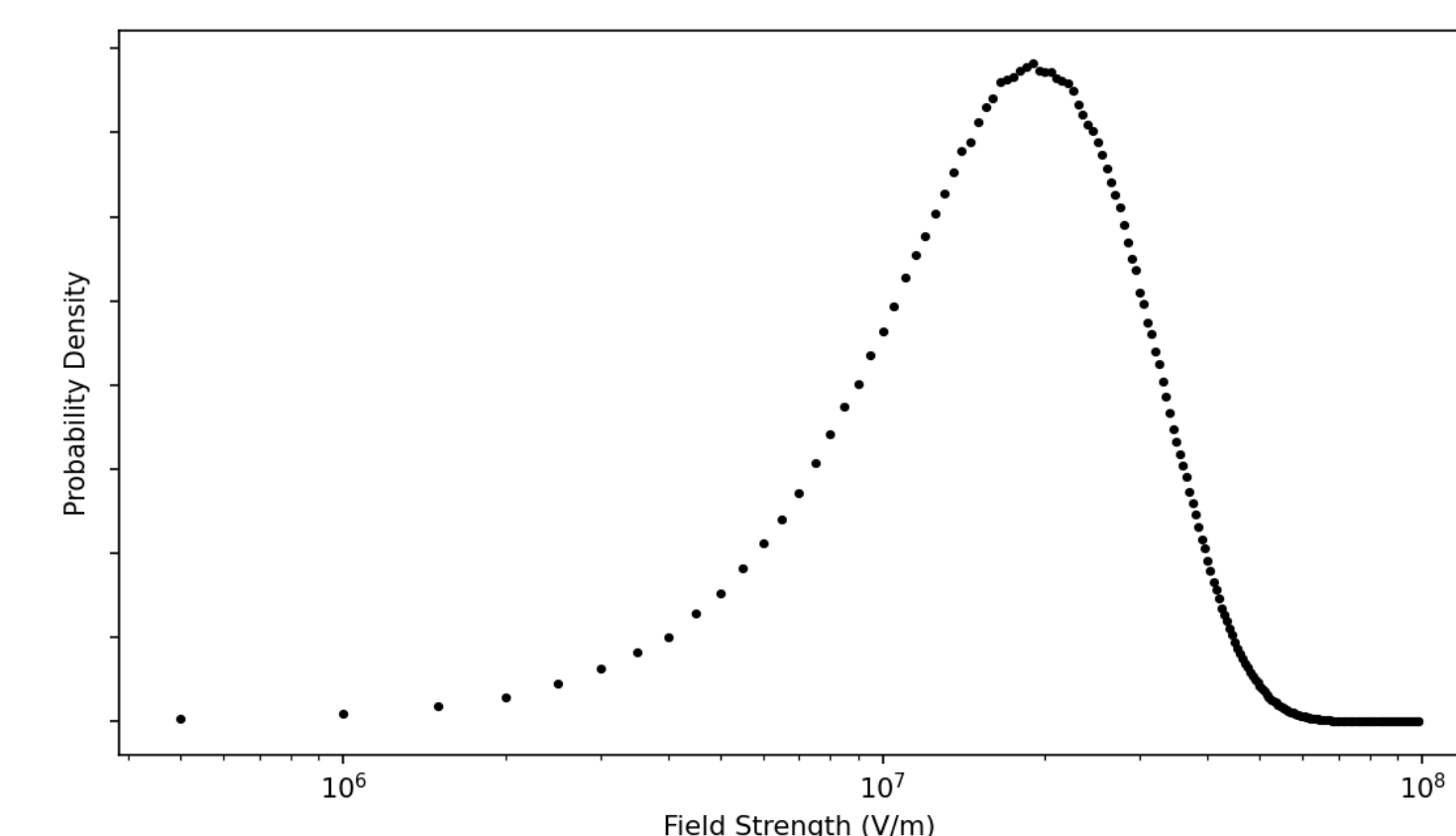
SUMMER RESEARCH

Abstract: Plasmas—fully ionized hot gases—can be found in many places in the universe. The way that energy is transported through astrophysical plasmas determines the size, brightness, and lifespan of stellar and compact objects. In order to most accurately model these systems, we need accurate models of their plasmas on an atomic level.

Methods: I use LLNL's molecular dynamics code, ddcMD, to simulate local, microscopic electric fields. Variations in these fields are known to affect the opacities, which are responsible for energy transport.



Top left: an example of an observed stellar spectrum. This gives astronomers information about the composition of a star or compact object. Top right: iron opacities at two different temperatures and densities. There is a discrepancy between the observed and calculated line shapes in the hotter, denser regime on the bottom. Left: an example of Stark broadening, where a line shape changes in the presence of an electric field. Below: a preliminary microfield distribution from ddcMD that demonstrates its capabilities.



Results: My microfield distributions will be used to calculate Stark broadening of these systems, which can then be used by external programs to calculate their line shapes. We hope to rectify the discrepancy in iron opacities at high temperatures and densities.

LLF AWARD

- I am incredibly grateful to be an LLF HEDS fellow. I appreciate that I have found a community already in my first few weeks thousands of miles from home.
- I will use the award to re-apply to PhD programs this fall. I also hope that the experience I've gained through my Master's program and as an intern at LLNL make me a promising candidate, so that I might earn my PhD and return to the lab as a postdoctoral research fellow!



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